

10. Analysis A: noise vs Z

The noise is split into three parts: temporal (frame-to-frame), fixed-pattern (static, tied to the projector pattern and pixel position), and total. The temporal part is fitted to a disparity-noise model, which yields the forward-model parameters.

1. Load and mask. Convert depth codes to mm with DEPTH_LSB_MM and set no-reads to NaN. Project the plate outline into the image using the registered pose. Shrink it by BOUNDARY_BAND_HALF_WIDTH_PX to form the region of interest (ROI).
2. Ground truth. For each ROI pixel, intersect its ray with the registered plate plane. This gives $Z_{GT}(u, v)$.
3. Temporal noise. For each pixel, compute $\sigma_t(u, v)$ as the standard deviation over frames (ddof = 1). Report the ROI median and 95th percentile for each station. Save the σ_t map, since its structure shows the projector-dot layout.
4. Fixed-pattern noise and bias. Average the frames to get $\bar{Z}(u, v)$. Fit a free plane to $\bar{Z}$; the standard deviation of the residual is σ_{fp} . The mean of $(\bar{Z} - Z_{GT})$ over the ROI is $\text{bias}(Z)$. Also report the angle between the fitted plane and the registered plane.
5. Total noise. For each frame, compute the residual from Z_{GT} . σ_{tot} is the RMS over the ROI and frames. Check that $\sigma_{tot}^2 \approx \sigma_t^2 + \sigma_{fp}^2 + \text{bias}^2$; a large mismatch points to an analysis error or to non-stationary noise.
6. Fill rate. Report the fraction of valid pixels in the ROI, for each station and tilt.
7. Spatial correlation. Compute the normalized 2-D autocorrelation of the temporal component ($Z - \bar{Z}$), averaged over frames, and of the fixed-pattern residual map. The correlation length along H and along V is the lag where the normalized autocorrelation first falls to 1/e. Compare it with the Tier-A IDW neighborhood N and with PHASE_JITTER_SPAN_PX; the span must exceed it.
8. Quantization. Histogram the raw depth codes for the central 20×20 px over all frames. The most common spacing between populated levels is $\delta Z_q(Z)$. If δZ_q equals DEPTH_LSB_MM at every Z, the output LSB is the quantizer and disparity quantization cannot be seen here; use the B-Z staircase instead. Otherwise compute $q = \delta Z_q \cdot k / Z^2$ at each Z. It should be constant if the quantizer works in disparity.
9. Noise model. Fit $\sigma_t(Z)$ by weighted least squares. The first term, σ_0 , is a floor that does not depend on Z (LSB/ $\sqrt{12}$ or electronics). The second is disparity noise σ_d propagated through $d =$

k/Z. Also fit a free power law $\sigma_t = a \cdot Z^n$. Triangulation predicts $n \approx 2$, and a clearly different n is a finding worth reporting.

$$\sigma_t(Z) = \sqrt{\sigma_0^2 + \left(\frac{\sigma_d Z^2}{k}\right)^2}$$

1. Incidence angle. From the tilt sub-series, plot σ_t , σ_{tot} , and fill rate against incidence angle at each Z.
2. Drift correction. Plot the sentinel mean Z against time and sensor temperature. If the drift exceeds $WARMUP_DRIFT_FRACTION_OF_SIGMA \times \sigma_t$, subtract a time-interpolated sentinel offset from the bias estimates and report the size of the correction.
3. Continuity with earlier data. Also compute the existing `testZRepeatabilityBrownBoard.py` metrics (boxes of half side `LEGACY_BOX_HALF_PX`: mean min–max range and mean per-pixel std) at the `LEGACY_METRIC_DEPTHS_MM` stations, which the A series adds to the ladder for this purpose. This lets the new results be compared directly with the existing VSX3000 data sets.
4. Outputs. `A_noise_summary.csv` with one row per station. Figures (PNG and SVG): the three σ curves vs Z with the model fit; fill rate vs Z; σ_t maps; autocorrelation profiles; depth-code histograms. The fitted σ_d , σ_0 , q, k, and correlation length go in `forward_model_parameters.json` for the Tier-A simulator.